

USER CENTERED DESIGN



CHAPTER Seven

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What is a user-centered approach?

User-centered approach is based on:



— Early focus on users and tasks:

- * directly studying cognitive,



- * behavioural,



- * anthropomorphic &

- * attitudinal characteristics



User-centered approach is based on:



– Empirical measurement:

* users' reactions and performance to

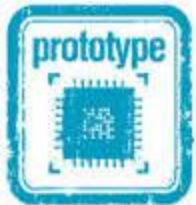


- scenarios,

- manuals,



- simulations & prototypes are observed,



recorded and analysed

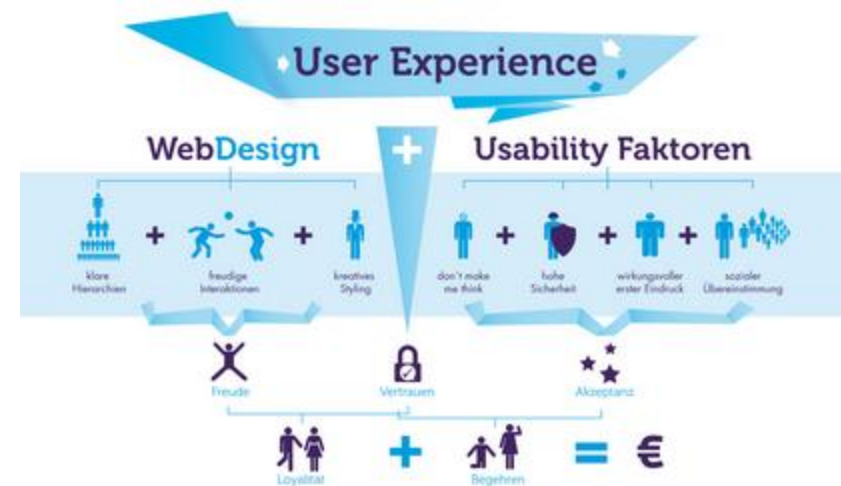


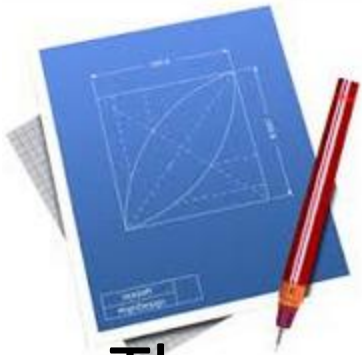
User-centered approach is based on:

- problems are found in user testing, fix them and carry out more tests



HOW





Software engineering

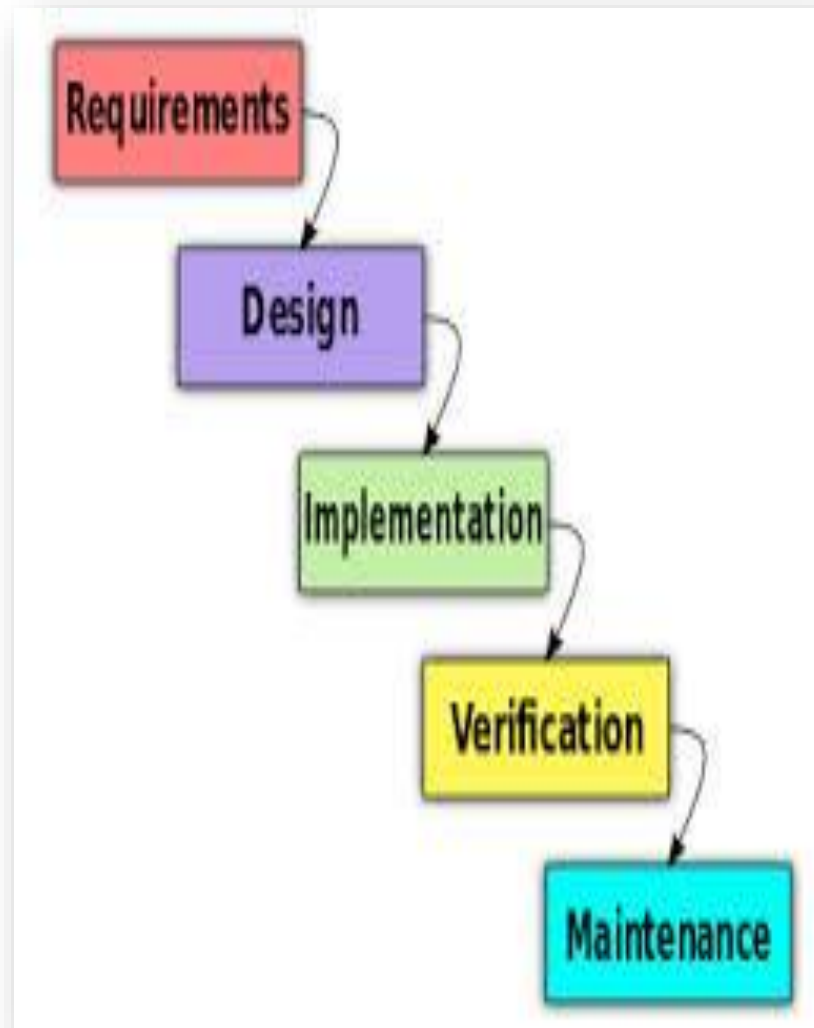
- The software engineering lifecycle helps to divide development into a number of different stages. Its usefulness is as a reference model and not as a detailed framework for project management. In many contexts, it may not be possible to follow all of the stages mentioned in exactly the order presented.

Software engineering

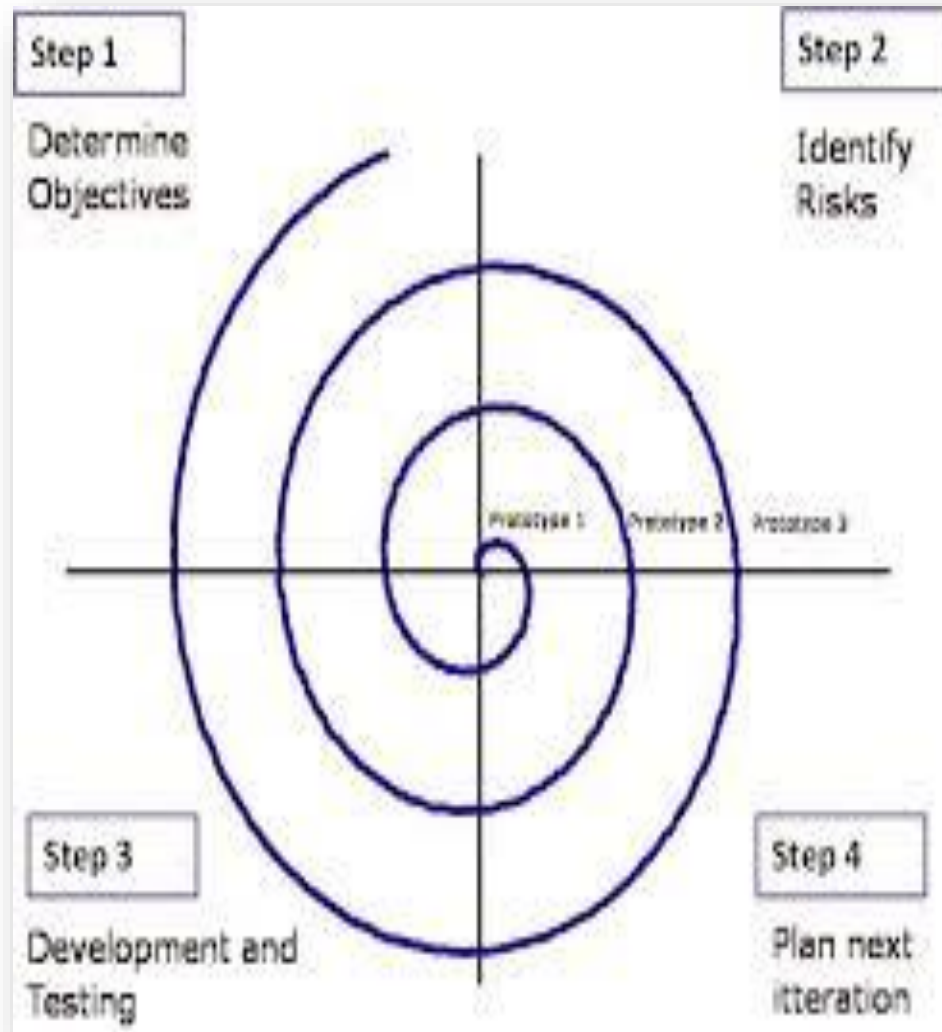
- Having said that, it does accurately describe many development projects and can be used to identify different activities during interface design.



The Software Engineering Life Cycle - water fall models



The Software Engineering Life Cycle - spiral models



The Software Engineering Life Cycle

- **Requirements elicitation** :This involves the identification of the various priorities that different parties will have for new pieces of software. Engineers must also identify the wider constraints that affect the system. The first half of this talk will cover the key techniques of interviewing and questionnaires.



The Software Engineering Life Cycle



- **Specification:** This involves the high level representation of a design. Key techniques include systems analysis and design methodologies as well as formal methods. The second half of this talk will explore the semi-formal QOC notation.

The Software Engineering Life Cycle



- **Design:** This involves the more detailed selection of particular options from amongst a range of alternative techniques. The next set of talks will discuss a wide range of dialogue styles that might be used during interface design.

The Software Engineering Life Cycle

- **Implementation:** Building the system to the required standard.



The Software Engineering Life Cycle

- **Testing:** This centres upon the evaluation of a system. The course will end with practical exercises in evaluation techniques.



The Software Engineering Life Cycle



- **Maintenance:** It is, typically, not enough to deliver a system. The results of testing may force modifications to an initial system. This also will be covered in the final stages of the course.

The Software Engineering Life Cycle

- The 'classic' software engineering lifecycle does not represent user interface design as a core activity. It has some role in requirements elicitation and testing. It is not clear whether the developers of such a model intended HCI to go on alongside systems engineering.



The Software Engineering Life Cycle

- This is the approach adopted by Boeing's concurrent engineering techniques. Or is it seen as a specialist activity that might occur after the implementation has been developed? Such unresolved questions have led a number of analysts to develop different models for HCI projects.



The Software Engineering Life Cycle

- It is critically important to decide where 'HCI' activities lie in the project structure. If they are relegated to the end of the development process then they are first in the firing line when budgets are cut and deadlines shortened.



The HCI Lifecycle.

- One of the problems with the traditional model for software development is that it does not clearly identify a role for HCI at any point in development.



The HCI Lifecycle.

- User interface concerns are 'mixed in' with wider development activities. This may result in one of two problems. Either HCI is ignored or it is relegated to an afterthought during the later stages of design. In either case, the consequences can be disastrous.



The HCI Lifecycle.

- If HCI is ignored then there is a good chance that problems will occur in the testing and maintenance stages. If HCI is relegated until late in the development cycle then it may prove very expensive to 'massage' application functionality into a form that can be readily accessed by the user.



The HCI Lifecycle.

- In either case the cost of introducing 'usability' issues will rise, the later you postpone it in the development cycle.



Design cycles of interactive system

- Design is a practical and creative activity with the aim of developing a product that helps its users achieve their goals.



Design cycles of interactive system

- Four basic activities for interaction design are:
 - **Establishing requirements** for the user experience
 - **Designing alternatives** that meet those requirements
 - **Prototyping** the alternative designs so that they can be communicated and assessed
 - **Evaluating** what is being built throughout the process and the user experience it offers



Design cycles of interactive system

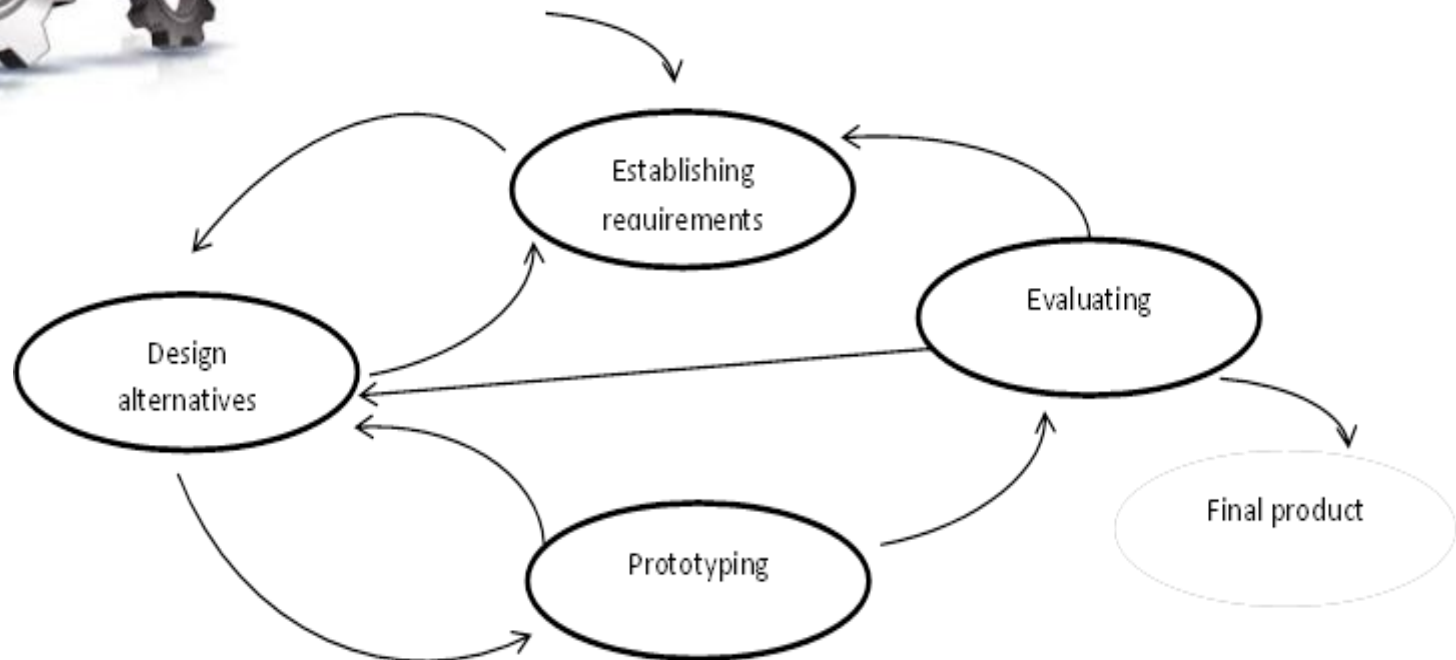


Figure 1 A simple interaction design life cycle

Design cycles of interactive system

- Establishing requirements



- In order to support people, we must know who our target users are and what kind of support an interactive product could usefully provide.



Design cycles of interactive system

- Establishing requirements



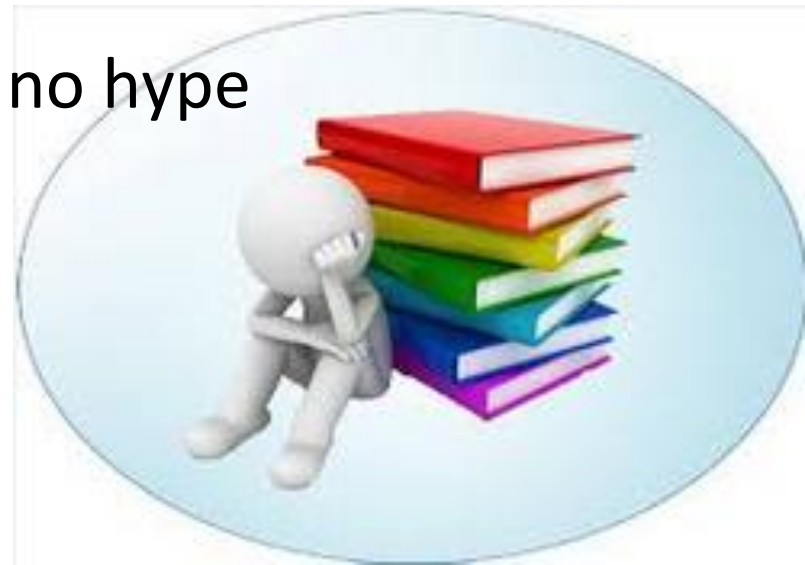
- These needs form the basis of the product's requirements and underpin the subsequent design and development. This activity is fundamental to a user-centered approach, and is very important in interaction design.



Understanding & Analyse the user

Importance of involving users

- Expectation management
 - Realistic expectations
 - No surprises, no disappointments
 - Timely training
 - Communication, but no hype



Understanding & Analyse the user

Importance of involving users

– Ownership

- Make the users active stakeholders
- More likely to forgive or accept problems
- Can make a big difference to acceptance and success of product



Understanding & Analyse the user

Degrees of user involvement

- Member of the design team
 - Full time: constant input, but lose touch with users
 - Part time: patchy input, and very stressful
 - Short term: inconsistent across project life
 - Long term: consistent, but lose touch with users



Understanding & Analyse the user

Degrees of user involvement

- Newsletters and other dissemination devices
 - Reach wider selection of users
 - Need communication both ways
- User involvement after product is released
- Combination of these approaches



Understanding & Analyse the user

Who are the users/stakeholders?

- Not as obvious as you think:
 - those who interact directly with the product
 - those who manage direct users
 - those who receive output from the product
 - those who make the purchasing decision
 - those who use competitor's products



Understanding & Analyse the user

Who are the users/stakeholders?

- Three categories of user (Eason, 1987):
 - **primary**: frequent hands-on
 - **secondary**: occasional or via someone else
 - **tertiary**: affected by its introduction, or will influence its purchase



Design cycles of interactive system

- Designing alternatives
 - This is the core activity of designing: actually suggesting ideas for meeting the requirements.

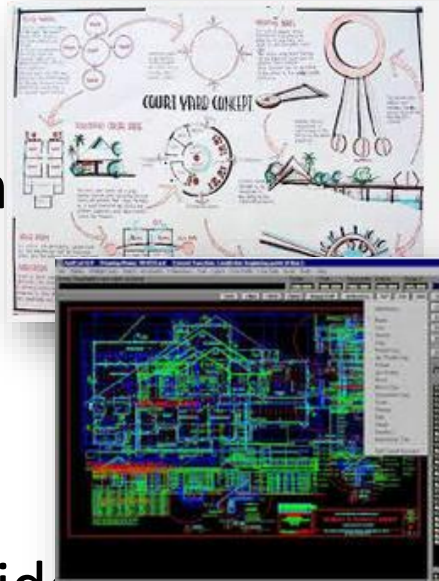


Design cycles of interactive system

- Designing alternatives
 - This activity can be broken up into two-sub activities:

1. conceptual design

1. physical design.



Alternatives are considered at every point.

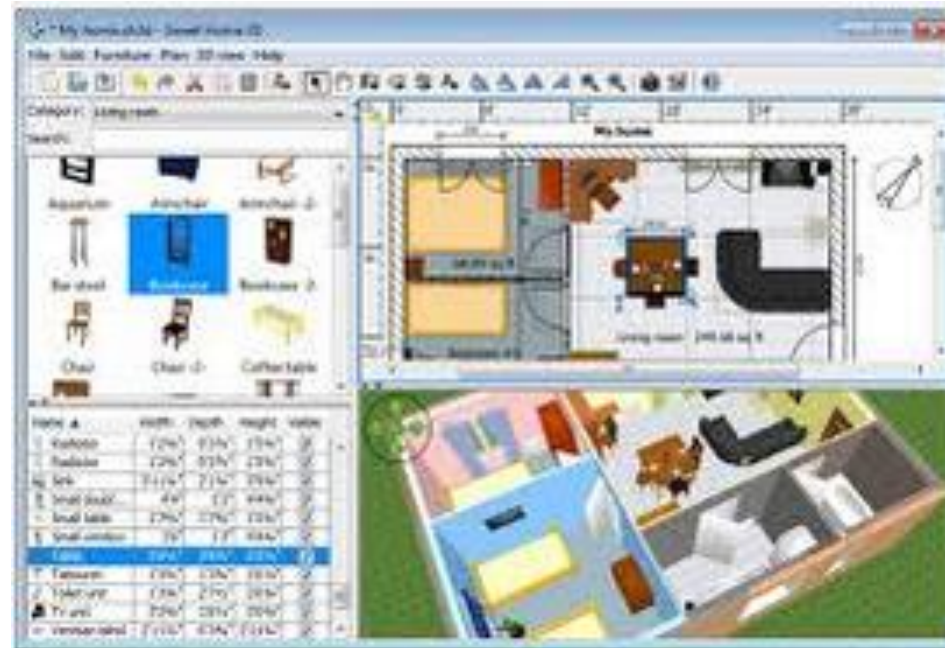
Design cycles of interactive system

conceptual design - involves producing the conceptual model for the product, and a conceptual model describes an abstraction outlining what people can do with a product and what concepts are needed to understand how to interact with it.



Design cycles of interactive system

physical design - considers the detail of the product including the colors, sounds and images to use, menu design, and icon design.



Design cycles of interactive system

- Prototyping
 - Interaction design involves designing interactive products. The most sensible way for users to evaluate such designs is to interact with them, and this can be achieved through the prototyping.



Design cycles of interactive system

- Prototyping
 - Does not necessarily mean a piece of software is required. There are different prototyping techniques, not all of which require a working piece of software.



Design cycles of interactive system

- Prototyping
 - For example, paper-based prototypes are very quick and cheap to build and are very effective for identifying problems in the early stages of design, and through role-playing users can get a real sense of what it will be like to interact with the product.



Prototyping

- A prototype is one manifestation of a design that allows stakeholders to interact with it and explore its suitability; it is limited in that a prototype will usually emphasize one set of product characteristics and de-emphasize others.



What is a prototype?

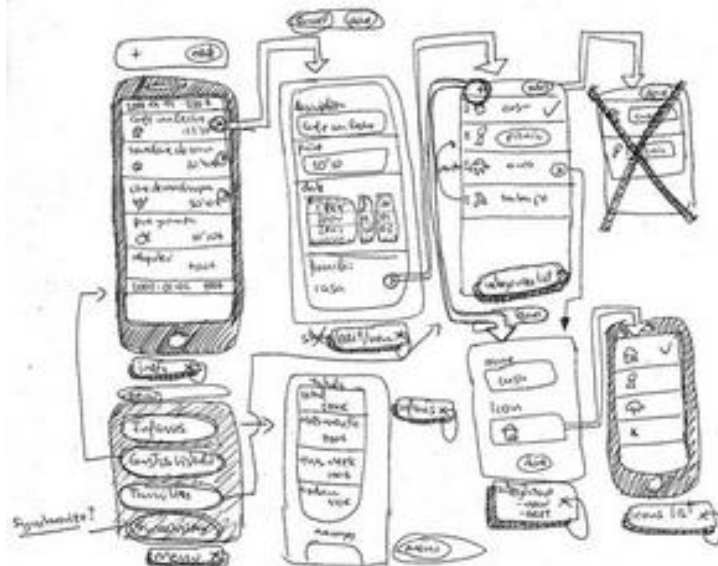
- In other design fields a prototype is a small-scale model:
 - a miniature car
 - a miniature building or town
 - the example here comes from a 3D printer



What is a prototype?

In interaction design it can be (among other things):

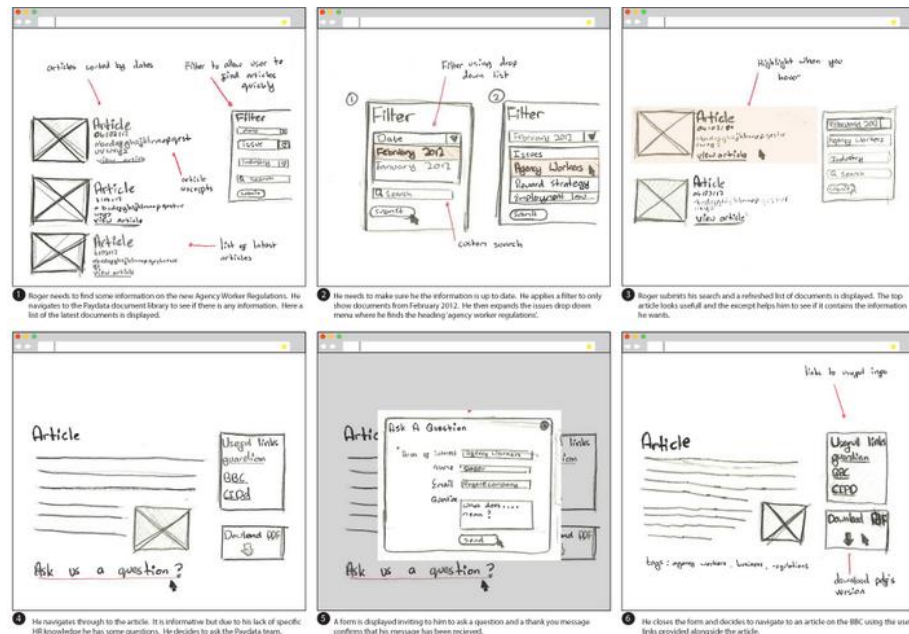
- a series of screen sketches



What is a prototype?

In interaction design it can be (among other things):

- a storyboard, i.e. a cartoon-like series of scenes



What is a prototype?

In interaction design it can be (among other things):

- a Powerpoint slide show



What is a prototype?

In interaction design it can be (among other things):

- a video simulating the use of a system



What is a prototype?

In interaction design it can be (among other things):

- a lump of wood (e.g. PalmPilot)



Figure 10: Reproduction of 3D "Sketch" of the PalmPilot
Very early on, Jeff Hawkins carved up a piece of wood to get a sense of the size that the device must be. It needed to fit into a shirt pocket, as well as sit comfortably in the palm of the hand. (Photo: Elizabeth Russ)

What is a prototype?

In interaction design it can be (among other things):

- a cardboard mock-up



What is a prototype?

In interaction design it can be (among other things):

- a piece of software with limited functionality written in the target language or in another language



Why prototype?

- Evaluation and feedback are central to interaction design
- Stakeholders can see, hold, interact with a prototype more easily than a document or a drawing
- Team members can communicate effectively



Why prototype?

- You can test out ideas for yourself
- It encourages reflection: very important aspect of design
- Prototypes answer questions, and support designers in choosing between alternatives



Prototyping

- When you here the term prototype, you may imagine something like a scale model of a building or a bridge, or maybe a piece of software that crashes every few minutes.



Prototyping

- A prototype can also be a paper-based outline of a screen or set of screens, an electronic picture, a video simulation of a task, or a three-dimensional paper and cardboard mock-up of a whole workstation.



Prototyping

- In fact a prototype can be anything from a paper-based storyboard through to a complex piece of software, and from a cardboard mock-up to a moulded or pressed piece of metal.



Filtering dimensions of prototyping



Filtering dimension	Example variables
Appearance	size; color; shape; margin; form; weight; texture; proportion; hardness; transparency; gradation; haptic; sound
Data	data size; data type (e.g., number; string; media); data use; privacy type; hierarchy; organization
Functionality	system function; users' functionality need
Interactivity	input behavior; output behavior; feedback behavior; information behavior
Spatial structure	arrangement of interface or information elements; relationship among interface or information elements – which can be either two-or three-dimensional, intangible or tangible, or mixed

Manifestation dimensions of prototyping



Manifestation dimension	Definition	Example variables
Material	Medium (either visible or invisible) used to form a prototype	Physical media, e.g., paper, wood, and plastic; tools for manipulating physical matters, e.g., knife, scissors, pen, and sandpaper; computational prototyping tools, e.g., Macromedia Flash and Visual Basic; physical computing tools, e.g., Phidgets and Basic Stamps; available existing artifacts, e.g., a beeper to simulate a heart attack
Resolution	Level of detail or sophistication of what is manifested (corresponding to fidelity)	Accuracy of performance, e.g., feedback time responding to an input by a user (giving user feedback in a paper prototype is slower than in a computer-based one); appearance details; interactivity details; realistic versus faked data
Scope	Range of what is covered to be manifested	Level of contextualization, e.g., website color scheme testing with only color scheme charts or color schemes placed in a website layout structure; book search navigation usability testing with only the book search related interface or the whole navigation interface

What to prototype?



- Technical issues
- Work flow, task design
- Screen layouts and information display
- Difficult, controversial, critical areas



Low-fidelity Prototyping

- Uses a medium which is unlike the final medium, e.g. paper, cardboard
- Is quick, cheap and easily changed
- Examples:



sketches of screens, task sequences, etc

‘Post-it’ notes

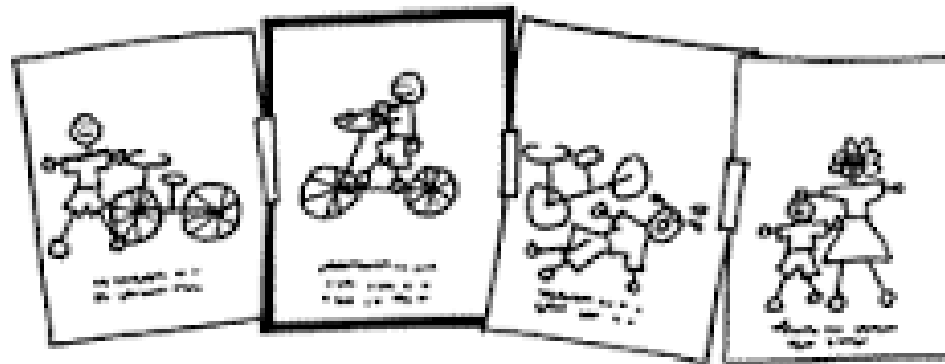
storyboards

‘Wizard-of-Oz’



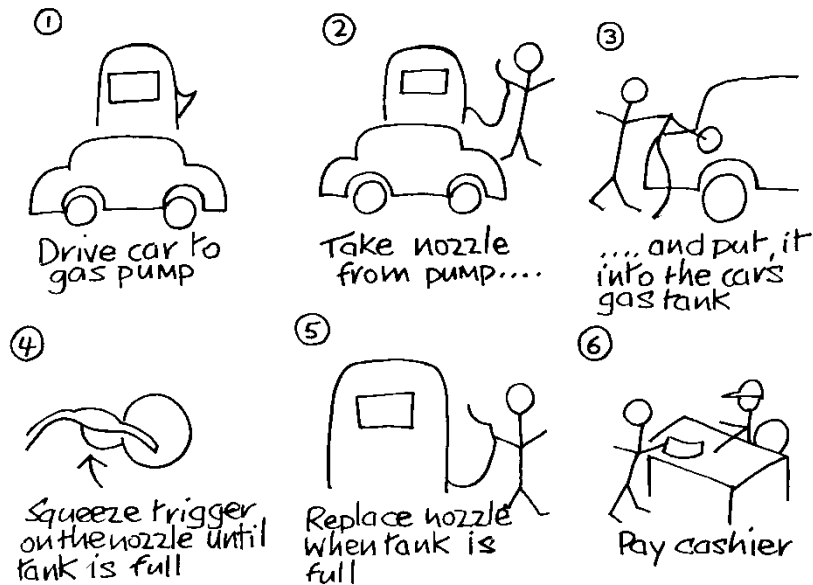
Storyboards

- Often used with scenarios, bringing more detail, and a chance to role play
- It is a series of sketches showing how a user might progress through a task using the device
- Used early



sketching

- Sketching is important to low-fidelity prototyping
- Don't be inhibited about drawing ability. Practice simple symbols



Card-based prototypes

- Index cards (3 X 5 inches)
- Each card represents one screen or part of screen
- Often used in website development



Travel

Organizer

23 August

WELCOME HELEN

Where do you want to go?

YORK

What date do you want to travel?

16 Sept

Which form of transport do you want?

TRAIN ▼

Do you need accommodation?

YES ▼



Travel
Organizer

23 August

Train timetable from Milton Keynes Central
to York
on 16 Sept

Depart	09:09	10:09	same	22:09
Arrive	12:30	13:30	past	01:30
			hour	

Accommodation	Hotel	B & B
	£40 to £150	£20 to £60

‘Wizard-of-Oz’ prototyping

- The user thinks they are interacting with a computer, but a developer is responding to output rather than the system.
- Usually done early in design to understand users’ expectations
- What is ‘wrong’ with this approach?



High-fidelity prototyping

- Uses materials that you would expect to be in the final product.
- Prototype looks more like the final system than a low-fidelity version.



High-fidelity prototyping

- For a high-fidelity software prototype common environments include Macromedia Director, Visual Basic, and Smalltalk.
- Danger that users think they have a full system.....see compromises



Compromises in prototyping

- All prototypes involve compromises
- For software-based prototyping maybe there is a slow response? sketchy icons? limited functionality?



Compromises in prototyping

- Two common types of compromise
 - ‘horizontal’: provide a wide range of functions, but with little detail
 - ‘vertical’: provide a lot of detail for only a few functions
- Compromises in prototypes mustn’t be ignored. Product needs engineering



construction



- Taking the prototypes (or learning from them) and creating a whole
- Quality must be attended to: usability (of course), reliability, robustness, maintainability, integrity, portability, efficiency, etc
- Product must be engineered
 - Evolutionary prototyping
 - 'Throw-away' prototyping

End of Lecture